Problem-solving practice 1

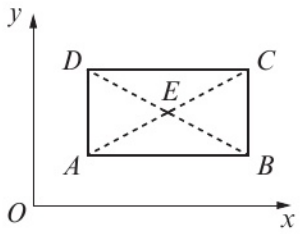
About half of the questions in your Foundation GCSE exam will require you to **problem-solve**, **reason**, **interpret** or **communicate** mathematically. If you come across a tricky or unfamiliar question in your exam you can try some of these strategies:

- ✓ Sketch a diagram to see what is going on.
- ✓ Try the problem with smaller or easier numbers.
- ✓ Plan your strategy before you start.
- ✓ Write down any formulae you might be able to use.
- ✓ Use x or n to represent an unknown value.

AO2 AO3

Now try this

1 In the diagram, $ABCD$ is a rectangle.



The length AB is twice the length AD.
E is the mid-point of the diagonals of the rectangle.
The coordinates of E are (30, 20).
Work out one set of possible coordinates for points A, B, C and D. **(4 marks)**

Coordinates page 36

There are lots of different correct answers to this question. Start by picking coordinates for A. In order for the width of the rectangle to be twice its height, the horizontal distance from A to E must be twice the vertical distance from A to E.

TOP TIP

Choose simple numbers to make your calculations easier. The coordinates of the mid-point are multiples of 10, so try using multiples of 10 for the coordinates of the vertices.

2 Dexter uses this formula to work out how long to cook a chicken.

Cooking time in minutes	=	chicken's weight in kg	× 45 + 30

He works out that his chicken will take exactly 1 hour 33 minutes to cook.
Work out the weight of Dexter's chicken. **(3 marks)**

Rearranging formulae page 50

The units in the formula are minutes, so write 1 hour 33 minutes as a time in minutes. To find the weight of Dexter's chicken, substitute any values you know into the formula, and then solve the equation to find the missing value.

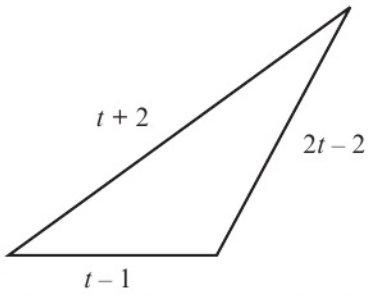
TOP TIP

You can represent the chicken's weight in kg as a single letter, for example w, in order to simplify your equation.

Problem-solving practice 2

Now try this

3



The perimeter of this triangle is 13 cm.
All lengths on the diagram are in cm.
Work out the value of ***t***. **(4 marks)**

4 Here are the first four terms of an arithmetic sequence:
9, 15, 21, 27, …
Explain why the number 65 cannot be a term of this sequence. **(3 marks)**

Worked solution video

5 A train travels a distance of y km in 3 minutes.
Show that the average speed of the train is $20y$ km/h. **(3 marks)**

6 A straight line has a gradient of 8 and passes through the point (5, 10).
(a) Determine whether the point (10, 50) lies on the line. You must show your working. **(3 marks)**
(b) Is your answer to part (a) reliable? Explain your answer. **(1 mark)**

Linear equations 2 page 31
Using algebra page 51

Use the information in the question to write an equation. Solve your equation to work out the value of t.

TOP TIP

Remember to simplify your equation by collecting like terms before solving.

Sequences 2 page 35

Work out the nth term of the sequence. Then work out consecutive terms of the sequence on either side of 65.

TOP TIP

You can use any successful strategy to answer a question, as long as you show your method. You could set the nth term equal to 65 and solve an equation to show that n is not an integer.

Formulae page 26 Speed page 64

You could convert minutes to hours then use the formula for average speed. Or you could find an expression for the distance travelled in 1 minute then multiply by 60.

TOP TIP

Try changing y into a number and calculating the average speed. It will give you an idea of how the formula works.

Straight-line graphs 2 page 39

You could tackle this question using algebra, or by drawing a diagram to scale. Whichever method you use, make sure you write a conclusion.

TOP TIP

If you have to comment on **reliability** then think about the method you used. If you used a diagram your answer might be less reliable than if you used algebra.